Ayomide Abilewa

Electronic and Electrical Engineering student, focused on embedded systems and instrumentation

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PROFILE

Works across embedded systems, instrumentation and computer vision. Has built sensing and detection systems from microcontroller nodes up to full-stack applications, and has taught electronics to secondary school students since 2023. Currently interning on Chevron Nigeria’s electrical and instrumentation team.

EDUCATION

B.Sc. Electronic and Electrical Engineering · Obafemi Awolowo University — 2021–2027 (expected)

Ile-Ife, Osun State, Nigeria

EXPERIENCE

Facilities Engineering Management Intern (SIWES) · Chevron Nigeria Limited — Mar 2025–Present

Lagos, Nigeria

- Rotate across the Electrical and Instrumentation (E&I) team, working on oil and gas field instrumentation and electrical systems.
- Interpret P&IDs and study control loop configurations, pneumatic signal standards (3–15 PSI), I/P converters and hazardous area classification.
- Study ESP systems, power transmission, single line diagram (SLD) interpretation and circuit breaker specifications.

Research and Development Intern (SIWES) · ACE Quanser Robotics Lab, Obafemi Awolowo University — 2024

Ile-Ife, Nigeria

- Operated the QDrone quadrotor UAV, QCar autonomous ground vehicle and QArm 6-DOF manipulator using MATLAB and Simulink.
- Applied PID control, sensor feedback and real-time system analysis to live hardware rather than simulation alone.
- Delivered a technical orientation presentation on the Quanser Aero 2 covering system architecture, control principles and laboratory safety.

Embedded Systems and EEE Tutor · Astar Tutorials — 2023–Present

Ile-Ife, Nigeria

- Teach circuit analysis, digital logic and microcontroller programming to engineering students.
- Design structured tutorial sessions, sequencing theory and hands-on exercises so students build working circuits.

PROJECTS

aniwe · Python, Streamlit, OpenRouter, Vision-language models — Aug 2026

- Shipped an AI costume-rating booth in Python and Streamlit with ten original hosts, per-session mood and quirk variation, and vision-language scoring across seven criteria collapsed into a single ranked score.
- Designed three-tier text-to-speech degradation (ElevenLabs, offline Piper, browser speech) and local faster-whisper speech-to-text so a blocked network reduces quality without stopping the system.

github.com/ayomide-abilewa/aniwe

AI-Powered Smart Attendance System · Python, OpenCV, YuNet, SFace — Feb–Mar 2026

- Owned the computer-vision pipeline and Flask REST API backend for an AI-powered attendance system using YuNet face detection and SFace recognition with cosine similarity matching.
- Developed the React, TypeScript and Tailwind frontend and a PostgreSQL schema covering students, courses, enrolment, sessions and attendance records.
- Defended the system at a final-year group project review and was invited to develop an extended production version.

github.com/ayomide-abilewa/Smart-Attendance-System

Live Video Pipe Anomaly Detection · Python, YOLOv8, YOLOv11, Weighted Boxes Fusion — 2026–Present

- Developing a two-stage cascaded detection architecture for real-time pipe anomaly identification, targeting industrial inspection.
- Assembled a parallel ensemble of RGB and edge-enhanced YOLO models merged with Weighted Boxes Fusion to improve precision across varied lighting and surface conditions.

Multispectral Acquisition Device for Hydroponics · ESP32, GY-2561, GY-31, I2C — 2025

- Built a multi-sensor ESP32 acquisition node measuring light intensity and colour spectrum with the GY-2561 and GY-31, streaming live to a Blynk IoT Cloud dashboard.
- Integrated multiple I2C sensors on a single ESP32, handling bus arbitration and sensor calibration.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, JavaScript, Verilog/HDL, Bash

Embedded and Hardware: Arduino, ESP32, STM32, Raspberry Pi, FPGA, I2C / SPI / UART, Sensor integration

AI and Machine Learning: YOLOv8/v11, OpenCV, TensorFlow (fundamentals), PyTorch (fundamentals), Weighted Boxes Fusion

Instrumentation and Control: P&ID interpretation, PID control loops, Pneumatic systems, I/P converters, 4–20 mA and 3–15 PSI standards

Electrical Systems: SLD interpretation, Circuit breaker sizing, ESP systems, Power transmission, Arc flash safety, Hazardous area classification

Tools: VS Code, Git/GitHub, MATLAB/Simulink, KiCad, Flask, React/TypeScript, PostgreSQL, Blynk IoT

LEADERSHIP AND TEACHING

SPAW Project Lead and Syllabus Team Lead · IEEE OAU Student Branch — 2024–Present

- Lead SPAW 3.0, a six-session embedded systems and entrepreneurship curriculum for secondary school students.
- Design all lesson plans, tutor guides and laboratory exercises for the programme.
- Produce structured PDF session materials including circuit diagrams, code examples and troubleshooting guides.

Core Instructor · Circuit Zero to Hero (ZTH) — 2023–Present

- Run practical circuit design and electronics workshops for students with no prior background in the subject.

TRAINING AND CERTIFICATIONS

Operational Excellence Training · Chevron — 2025

Cyber Security and Data Privacy Awareness · Chevron — 2025

Business Conduct, Ethics and Conflict of Interest · Chevron — 2025

FPGA Design and HDL/Verilog Workshop (SWEP) · Faculty of Engineering, Obafemi Awolowo University — 2024

Robotics Lab Orientation and Operations · ACE Quanser Lab, Obafemi Awolowo University — 2024